

WEEK 6: FUNCTIONAL ANATOMY & KINESIOLOGY

BACK & SPINE

- Vertebral spine or vertebrae
 - 33 in children; 26 in adults
 - Sacral & coccyx in adult (1+1 instead of 5+4)
- Curves of the spine
- Parts of vertebrae (body, transverse and spinous process, transverse foramen, pedicle anterior, lamina posterior, vertebral arch)

VERTEBRAE

TYPICAL VERTEBRA

- Cervical (C3-6); thoracic (T5-8); lumbar (L1-4)

Feature	Cervical	Thoracic	Lumbar
Vertebral body	Small	Moderate sized, heart shaped	Large/ massive, kidney shaped
Spinous process	Bifid spinous process, short	Long and inclined downward/ almost vertical	Short, Thick, Flat, quadrilateral
Transverse process	Transverse foramen	Costal Facets/ pits	Conical shape
Spinal canal/ Vertebral foramen	Largest and triangular	Smallest and circular	Triangular
Facets	SAF: Posterior/ Superior IAF: Anterior/ Inferior	SAF: Posterior/ Lateral IAF: Anterior/ Medial	SAF: Medial IAF: Lateral
Angle	45°	60°	90°

- **Transverse foramen** - vertebral artery for circle of Willis

ATYPICAL VERTEBRA

1.) Cervical

- C1 (atlas): no vertebral body & spinous process
- C2 (axis): dens/odontoid process (attachment to atlas); short transverse process; large, bifid spinous process; largest spinous process
- C7 (vertebral prominens): long, non-bifid spinous process; longest spinous process

2.) Thoracic

- T1: horizontal spinous process; entire facet above, demifacet below; large costal facet in rib 1
- T2-4: slightly atypical due to small size
- T9: tubercles; demifacet above
- T10: single facet
- T11: single facet; no facet on transverse process

- T12: one entire facet; no facet on transverse process; inferior articular process convex & lateralward

1.) T1

- Special because it is a transition between cervical and thoracic spine
- Unique features:
 - Full superior costal facet for rib 1
 - Demifacet inferiorly for rib 2
 - Body is more similar to cervical vertebra (smaller than lower thoracic)
 - Spinous process is long but less sharp than mid-thoracic
- Dysfunction in transition leads to upper thoracic stiffness, cervicothoracic junction pain, and thoracic outlet syndrome (relation with rib 1)

2.) T9

- Partially atypical because it only has a superior demifacet and inferior demifacet may be absent
 - Rib articulation pattern starts to change

3.) T10

- Single whole facet on each side (not demifacets)
- Has transverse process

3.) Lumbar

- L5: stout transverse process

JOINT COMPLEXES

Cartilaginous joint for stability and integrity of IV discs

1.) 3 joint complex

- Adjacent vertebral bodies and IV discs (1)
- Paired superior and inferior facets; between pedicle and lamina (2)

2.) 5 joint complex — cervical uncovertebral joints or joints of Von Luschka

- Vertebral body of superior vertebra and paired base of pedicles of inferior vertebra
- Formed by uncinate processes of the vertebra below interlocking with the vertebral bodies above
- Unique to cervical spine between C3-7
- Most common site of degeneration (arthritis & bone spurs)

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JOINTS

1.) **AO joint** - yes joint

2.) **AA joint** - no joint

1.) **50% of joint movement - AO joint**

- C3-C7 (C5-C6) - flexion

2.) **50% of joint movement - AA**

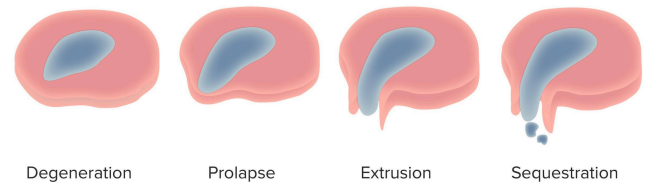
- C3-7 (C6-7) - rotation

uhm.. pero based sa google, 50% are both AA and AO, then remaining is movement ng C3-5 comprising the 30%

Feature	ANNULUS FIBROSUS	NUCLEUS PULPOSUS
Composition	A fibrocartilaginous tissue consisting of layers of lamellae rich in highly cross-linked collagen fibrils	More amorphous with a small percentage of randomly oriented fibrils and a significant content of proteoglycans
Structure	The tough circular exterior of the intervertebral disc surrounding the soft inner core, the nucleus pulposus	The soft, gelatinous central portion of the intervertebral disk that moves within the disk with changes in posture
Location	Occur in the exterior of the intervertebral disc	Occur interior of the intervertebral disc
Proteoglycan	Contains 20% proteoglycan	Contains 65% proteoglycan
Collagen	Contains 60% collagen	Contains 20% collagen
Water	Contain less water	80-90% water
Function	Enhances the spine's rotational stability and helps resist compressive stress	Allows the spine to absorb shock

- Nucleus - 80-90% water

MCNAB'S CLASSIFICATION OF HERNIATION



Feature	Degeneration	Prolapse (Protrusion)	Extrusion	Sequestration
Definition	Symmetrical outward extension of disc beyond vertebral margin	Focal herniation where nucleus pushes into annulus but outer fibers still intact	Nucleus pulposus breaks through annulus but remains connected to main disc	Free fragment of nucleus completely separates from main disc
Annulus Fibrosus Status	Intact	Inner fibers disrupted, outer fibers intact	Completely ruptured	Completely ruptured
Nucleus Pulposus	Contained within annulus	Displaced but still contained	Escapes beyond annulus	Detached fragment migrates
Shape	Broad-based	Localized protrusion	Narrow neck with wider herniated portion	Free fragment in canal
Disc Containment	Contained	Contained	Non-contained	Non-contained

- Reversible - degeneration and prolapse (more intense intervention)

- Slip disc - degeneration & prolapse
- Herniated disc - extrusion & sequestration

- Posterolateral bulging due to ALL and PLL orientation

- Schmor's nodes - fragments from sequestration

- Prolapse can already present sensory problem (sharp-shooting pain)

PRESSURE ON IV DISCS

IMPORTANT LANDMARKS

1.) **Cervical**

- C3 - hyoid
- C4 - thyroid
- C6 - cricoid (back of the neck; suboccipital)

2.) **Thoracic**

- T2 - superior angle of scapula
- T3 - scapular spine
- T7 - inferior angle
- T10 - xiphoid process (dermatome is T6)

3.) **Lumbar**

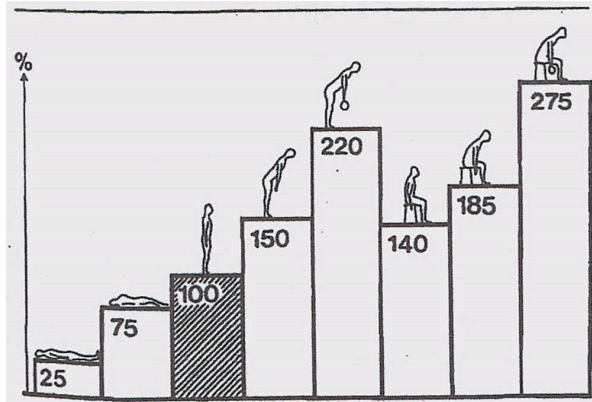
- L4 - iliac crest
- L5 - tubercle of iliac crest
- S2 - sacroiliac and PSIS

IV DISCS

- Located between adjacent vertebral bodies
- Forms part of the 3-joint complex
- Account for 25% of total vertebral column height (1:4 ratio)
- Function: shock absorption, load transmission, spinal mobility

COMPOSITION

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- 1.) 25% - supine
- 2.) 75% - sidelying
- 3.) 100% - standing upright
- 4.) 140% - sitting upright
- 5.) 160% - standing + bending forward
- 6.) 185% - sitting + bending forward
- 7.) 220% - standing + bending forward + weights
- 8.) 275% - sitting + bending forward + weights

BIOMECHANICS OF SPINE

- Proximate - dekat; separate - hiwalay

Biomechanics	Vertebral body	Spinous process	IV Discs	Foramina	Facets
Flexion	Approximates	Separates	Flattens (Ant)	Opens	Opens, anterosuperior movement
			Pushed (Post)		
Extension	Separates	Approximates	Flattens (Post)	Closes	Closes, Posteroinferior
			Pushed (Ant)		
Lateral Flexion	Approximates (Ipsi)	-	Flattens (Ipsi)	Closes (Ipsi)	Closes (Ipsi)
	Separates (Contra)		Pushed (Contra)	Opens (Contra)	Opens (Contra)
Rotation	Rotates (Ipsi)	Rotates (Contra)	At its weakest	Closes (Ipsi)	Closes (Ipsi)
				Opens (Contra)	Opens (Contra)
Distraction	Separates	-	-	-	-
Compression	Approximates	-	-	-	-
Shearing	Sliding ant, post, lat	-	-	-	-

ROM OF SPINE

Motion	AO joint	AA joint	C3-C7	Thoracic	Lumbar
Flexion	10°	5°	45°	15°	40°
Extension	25°	10°	45°	15°	25°
Lateral flexion	L & R 5°	L & R 10°	L & R 30°	L & R 15°	L & R 20°
Rotation	L & R 0°	L & R 45°	L & R 30°	L & R 40°	L & R 5°
	50% flex/ ext	50% rot			

- Sing Voltes V rhythm

- In the end: Yan lahat ang movement ng spine kaya dapat nasa mind
- Ang cervical (cervical) at thoracic (thoracic) pati lumbar spine

SPINAL LIGAMENTS

INTERSEGMENTAL LIGAMENTS

1.) Connect many vertebrae

Ligament	Attachment	Taut	Lax
Anterior Longitudinal Ligament (ALL)	Connects anterior aspect of vertebral bodies	Extension	Flexion
Posterior Longitudinal Ligament (PLL)	Connects posterior aspect of vertebral bodies (inside vertebral canal)	Flexion	Extension
Supraspinous Ligament	Connects tips of spinous processes (C7 to Sacrum)	Flexion	Extension

2.) Connect individual or adjacent vertebrae

Ligament	Attachment	Taut	Lax
Ligamentum Flavum (Flava)	Connects laminae of adjacent vertebrae	Flexion	Extension
Ligamentum Nuchae	Connects spinous processes from Occiput to C7	Flexion	Extension
Interspinous Ligament	Connects adjacent spinous processes	Flexion	Extension
Intertransverse Ligament	Connects adjacent transverse processes	Contralateral lateral flexion	Ipsilateral lateral flexion

- Interspinous - pairing of spinous process

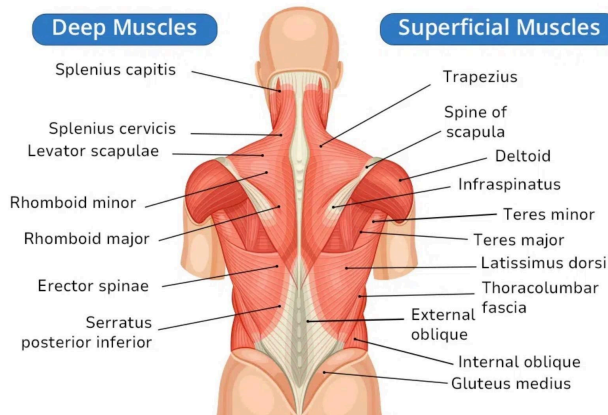
3.) Other important cervical ligaments

Ligament	Function	Clinical Correlation
Alar Ligaments	Limit rotation of skull and atlas over axis ("wing ligaments")	Excessive rotation injury
Transverse Ligament	Holds dens against anterior arch of atlas	Jefferson's fracture (C1), Hangman's fracture (C2)
Cruciform Ligament	Vertical ligament over transverse ligament; limits skull & atlas rotation	Upper cervical instability
Accessory Atlanto-Axial Ligament (AAA Lig.)	Assists alar ligament in limiting rotation	Upper cervical stability

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MUSCLES OF THE BACK

Diagram of Back Muscles



CLINICAL SIGNIFICANCE

- 1.) Narrow pedicle - due to extra bone formation causes sensory problems
- 2.) Spinal stenosis - back pain, pain in walking downhill + leg pain (calf)
- 3.) Scoliosis - problem in the vertebral body
 - Levo - left curve
 - Dextro - right curve
- 4.) Osteoporosis - most common in adult
- 5.) Dowager's hump - hyperkyphosis of thoracic
- 6.) Slip disc - problem in body (specifically in ligaments; ALL and PLL)
- 7.) Radiating pain and weakness - problem in pedicle and transverse foramen
- 8.) Hoop stretch - extra stretch in iv disc; compressed annulus causes tear (nucleus is out)
- 9.) Ergonomic bulging - L4-5